File permissions in Linux

Project description

I am a security professional at a large organization. I mainly work with their research team. Part of my job is to ensure users on this team are authorized with the appropriate permissions. This helps keep the system secure.

My task is to examine existing permissions on the file system. I will need to determine if the permissions match the authorization that should be given. If they do not match, I will need to modify the permissions to authorize the appropriate users and remove any unauthorized access.

Check file and directory details

To check the file and directory details of the folder I am currently located in, I use the following commands:

```
researcher2@30c64fecb119:~$ pwd
/home/researcher2
researcher2@30c64fecb119:~$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 12 15:56 .
drwxr-xr-x 1 root        root          4096 Jun 12 15:01 ..
-rw------- 1 researcher2 research_team   10 Jun 12 15:57 .bash_history
-rw-r--r-- 1 researcher2 research_team  220 Mar 27  2022 .bash_logout
-rw-r--r-- 1 researcher2 research_team 3574 Jun 12 15:02 .bashrc
-rw-r--r-- 1 researcher2 research_team 3574 Jun 12 15:02 .profile
drwxr-xr-x 3 researcher2 research_team 4096 Jun 12 15:02 projects
```

The **ls -la** command shows me every file, including the hidden ones, that are in the current folder. Another alternatives are **ls -l** which shows every file without the hidden ones in a long format, and **ls -a** that shows every file including the hidden ones in a normal short format. The command I have used is a combination of the two explained.

Describe the permissions string

I will take for example the permissions string for the projects directory: **drwxr-xr-x**

The first symbol indicates whether an item is a directory or a file. The ‘d’ shows that it is a directory. For files, normally it is ‘-’.

The symbols 2-4 show the permissions of the owner of the item, e.g. user permissions. In this case the user has permissions to read, write and execute (open) the current folder.

The next three symbols indicate the permissions of the user's group. Here, the group can only read and open the directory's content, without creating or modifying anything.
The last three symbols indicate the permissions of others. These include users and groups, different from the ones given, that are available in the system. In this case, they have the same permissions as the group -> read and execute.

Change file permissions

To change the file permissions, I use the following command:

```
researcher2@30c64fecb119:~$ chmod g+w projects/
researcher2@30c64fecb119:~$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 12 15:56 .
drwxr-xr-x 1 root        root          4096 Jun 12 15:01 ..
-rw------- 1 researcher2 research_team   63 Jun 12 16:06 .bash_history
-rw-r--r-- 1 researcher2 research_team  220 Mar 27  2022 .bash_logout
-rw-r--r-- 1 researcher2 research_team 3574 Jun 12 15:02 .bashrc
-rw-r--r-- 1 researcher2 research_team 3574 Jun 12 15:02 .profile
drwxrwxr-x 3 researcher2 research_team 4096 Jun 12 15:02 projects
```

The **chmod** command allows me to change a file or a directory's permissions, **chmod g+w** indicates that I want to add **w** (write) permission to the **g** (group) of the owner.
Now the owner's group has write permissions on the projects folder.

Change file permissions on a hidden file

```
researcher2@30c64fecb119:~$ cd projects/
researcher2@30c64fecb119:~/projects$ ls -la
total 32
drwxrwxr-x 3 researcher2 research_team 4096 Jun 12 15:02 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 12 15:56 ..
-rw--w---- 1 researcher2 research_team   46 Jun 12 15:02 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 12 15:02 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Jun 12 15:02 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jun 12 15:02 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun 12 15:02 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun 12 15:02 project_t.txt
```

In the projects folder, there is a hidden file, called ".project_x.txt". Hidden files and directories in my organization's system should not be overwritten or changed by anyone. So i changed the permissions of the file, so that only the user and the group can read it, without modifying it:

```
researcher2@30c64fecb119:~/projects$ chmod u-w,g-w .project_x.txt
researcher2@30c64fecb119:~/projects$ ls -la
total 32
drwxrwxr-x 3 researcher2 research_team 4096 Jun 12 15:02 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 12 15:56 ..
-r-------- 1 researcher2 research_team   46 Jun 12 15:02 .project_x.txt
```

Change directory permissions

Currently, the drafts directory in the projects folder has wrong permissions. Only the user should be allowed to access the drafts directory and its contents. This means that only the user should have execute privileges:

```
researcher2@30c64fecb119:~/projects$ chmod g-x drafts/
researcher2@30c64fecb119:~/projects$ ls -la
total 32
drwxrwxr-x 3 researcher2 research_team 4096 Jun 12 15:02 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 12 15:56 ..
-r-------- 1 researcher2 research_team   46 Jun 12 15:02 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Jun 12 15:02 drafts
```

Summary

Keeping correct permissions on every file and directory in a Linux system is important because permissions control **who can read, modify, execute, or delete resources**. In an organization, improper permissions can lead to security breaches, data loss, compliance violations, and operational problems.